TECHNICAL PROJECT REPORT

Design, Implementation and Validation of a Secure Enterprise Wireless Network

A Cisco Packet Tracer simulation



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Abstract

This report presents the design, implementation and validation of a secure enterprise wireless network built with Cisco Packet Tracer. Four departments (Administration, IT, Human Resources and Guest) share a three-tier hierarchical infrastructure made of a Cisco 2911 router, a Cisco 3560 Layer 3 switch, four Cisco 2960 access switches and four wireless access points.

Traffic is segmented with VLANs, routed between VLANs through switch virtual interfaces (SVIs) on the core switch, and controlled by two named extended ACLs that apply the principle of least privilege. Shared DHCP, DNS and HTTP services are hosted in the IT VLAN and reached through DHCP relay, while each department has its own WPA2-PSK SSID mapped to its VLAN.

A three-phase test campaign (baseline connectivity, service availability, security enforcement) confirmed that every requirement is met: Guest and HR traffic toward restricted networks is blocked, authorized traffic is unaffected, and round-trip times stay below 65 ms. The report also analyses the limitations of the design and lists the hardening steps required before a production deployment.

Keywords: VLAN, inter-VLAN routing, Layer 3 switching, WPA2, access control lists, DHCP relay, Rapid PVST+, Cisco Packet Tracer.

1. Introduction

Enterprise networks must deliver connectivity to several groups of users while keeping their traffic separated and protected. Wireless access adds a further challenge: the same infrastructure must serve staff and visitors without exposing internal resources to the latter.

This project designs a hierarchical enterprise network in which four departments operate on separate VLANs. It covers inter-VLAN routing on a Layer 3 switch, centralized DHCP, DNS and HTTP services, WPA2-protected wireless networks, and ACL-based segmentation of traffic flows. The network is built and tested entirely in Cisco Packet Tracer.

The report is organized as follows: Section 2 states the objectives, Section 3 describes the architecture, Section 4 the VLAN and IP addressing, Section 5 the configuration of each component, Section 6 the test campaign, Section 7 the security analysis, and Sections 8 and 9 the limitations, future work and conclusion. Device credentials in all listings are redacted.

2. Project Objectives

The project pursues the following objectives:

- Design a realistic enterprise topology with Cisco equipment (2911 router, 3560 Layer 3 switch, 2960 access switches, wireless access points).
- Implement VLAN segmentation to isolate departmental traffic and enforce security boundaries.
- Configure inter-VLAN routing on a Layer 3 switch to allow controlled communication between departments.
- Deploy centralized DHCP, DNS and HTTP services reachable from every authorized VLAN. • Secure wireless access with WPA2-PSK authentication and a distinct SSID per department.
- Apply ACL-based security policies to restrict Guest access and isolate sensitive departments.
- Validate all network services through systematic ping tests, DHCP verification and web browsing simulation.

Table 1. Requirements and how they are met

ID	Requirement	Implementation	Verified in
R1	Hierarchical topology with Cisco devices	R1 (2911), SW-CORE (3560), 4 x 2960, 4 access points	Section 3
R2	Isolation of departments	VLANs 10, 20, 30, 40 and management VLAN 99	Sections 4, 6.1
R3	Controlled inter-VLAN communication	SVIs on SW-CORE with ip routing	Section 6.1
R4	Centralized DHCP, DNS and HTTP	Servers in VLAN 20, DHCP relay on SVIs	Sections 6.1, 6.2
R5	Secure wireless access	WPA2-PSK with AES, one SSID per department	Section 5.8
R6	Policy-based traffic control	Named extended ACLs BLOCK_GUEST and BLOCK_RH_ADMIN	Section 6.3
R7	Systematic validation	Baseline, service and security test phases	Section 6

3. Network Architecture

3.1 Topology overview

The network follows a three-tier hierarchical model composed of edge, core/distribution and access layers. This organization provides scalability, a clear separation of responsibilities and a predictable place to apply policy.

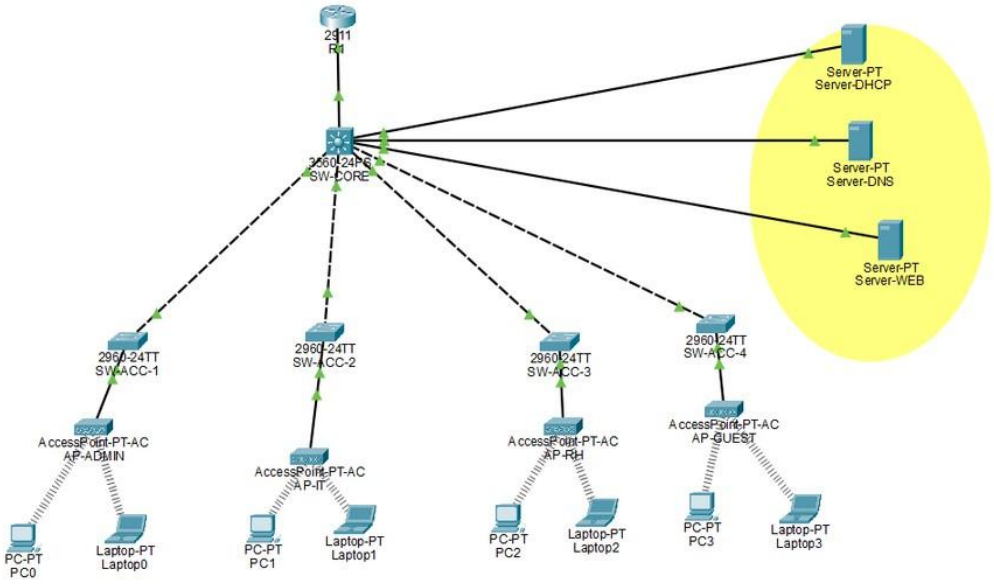


Figure 1. Network topology: router R1, core switch SW-CORE, four access switches with their access points, and the three servers in VLAN 20.

Table 2. Network tiers and device roles

Tier	Device	Role
Edge	Cisco 2911 router (R1)	Gateway toward the external side; static routes back to every VLAN

Core / Distribution	Cisco 3560-24PS (SW-CORE)	Inter-VLAN routing, trunk aggregation, DHCP relay, ACL enforcement, STP root bridge

Tier	Device	Role
Access	Cisco 2960 x 4 (SW-ACC-1 to SW-ACC-4)	Department switching and VLAN assignment
Wireless	Access points x 4 (AP-ADMIN, AP-IT, AP-RH, AP-GUEST)	One SSID and one VLAN per department
Servers	Server-DHCP, Server-DNS, Server-WEB	Centralized enterprise services in VLAN 20

3.2 Design rationale

Dedicated access points. In enterprise environments, wireless access is provided by specialized access points while routing, DHCP, DNS and security are handled by dedicated network equipment. This separation of responsibilities produces a design that is more scalable, more secure and easier to manage than one based on consumer Wi-Fi routers. Each access point is connected to its own access switch on a dedicated VLAN, which isolates wireless traffic from the other departments from the first hop.

Layer 3 core. SVIs on SW-CORE are the default gateways of all VLANs, so inter-VLAN traffic is routed at the core and ACLs can be attached directly to the VLAN interfaces.

Centralized services. A single DHCP server serves all VLANs through relay agents; DNS and the intranet web server are hosted beside it in VLAN 20.

3.3 Physical links

Table 3. Links between devices

Link	From	To	Type
Uplink	R1 Gi0/0 (10.0.0.1/30)	SW-CORE Gi0/1 (10.0.0.2/30)	Routed point-to-point
Trunks	SW-CORE Fa0/1 to Fa0/4	SW-ACC-1 to SW-ACC-4	802.1Q, VLANs 10, 20, 30, 40, 99
Servers	SW-CORE access ports	DHCP, DNS and WEB servers	Access ports, VLAN 20
Wireless	SW-ACC-x Fa0/2	Department access point	Access port, VLAN 10, 20, 30 or 40

4. IP Addressing and VLAN Design

4.1 VLAN table

The network is divided into five VLANs, each mapped to one /24 subnet whose default gateway is the SVI of that VLAN on SW-CORE.

Table 4. VLAN plan

VLAN	Name	Department	Network	Default gateway
10	ADMIN	Administration	192.168.10.0/24	192.168.10.1
20	IT	IT and shared servers	192.168.20.0/24	192.168.20.1

30	RH	Human Resources	192.168.30.0/24	192.168.30.1
40	GUEST	Guest Wi-Fi	192.168.40.0/24	192.168.40.1
99	MANAGEMENT	Device management	192.168.99.0/24	192.168.99.1

VLAN segmentation brings four main benefits. **Security:** departmental traffic is isolated and unauthorized inter-VLAN access is blocked by ACLs. **Performance:** smaller broadcast domains reduce unnecessary traffic. **Broadcast control:** each VLAN is its own broadcast domain, which limits broadcast storms. **Administration:** each VLAN can be managed and monitored independently.

4.2 Server addressing

All servers are hosted in VLAN 20 and are reached from the authorized VLANs through inter-VLAN routing. DHCP requests from other VLANs reach the server through relay.

Table 5. Server addressing

Server	IP address	Mask	Gateway	Service
Server-DHCP	192.168.20.10	255.255.255.0	192.168.20.1	DHCP for all VLANs
Server-DNS	192.168.20.20	255.255.255.0	192.168.20.1	Name resolution
Server-WEB	192.168.20.30	255.255.255.0	192.168.20.1	HTTP intranet portal

4.3 Wireless networks

Each department has a dedicated SSID configured on its own access point. All wireless networks use WPA2-PSK authentication with AES encryption. Pre-shared keys are not published in this report.

Table 6. SSID configuration

Access point	SSID	Security	Encryption	VLAN
AP-ADMIN	CORP_ADMIN	WPA2-PSK	AES	10
AP-IT	CORP_IT	WPA2-PSK	AES	20
AP-RH	CORP_RH	WPA2-PSK	AES	30
AP-GUEST	GUEST_WIFI	WPA2-PSK	AES	40

5. Configuration Details

5.1 SW-CORE: VLANs, SVIs and routed uplink

All VLANs are created on SW-CORE and an SVI is configured as the default gateway of each one. The command **ip routing** enables Layer 3 forwarding between VLANs. The uplink toward R1 is a routed port, and each access switch is reached over an 802.1Q trunk that carries every departmental VLAN.

```
hostname SW-CORE enable secret
<REDACTED> service password-
encryption vlan 10  name ADMIN vlan
20  name IT vlan 30  name RH vlan 40
name GUEST vlan 99  name MANAGEMENT
ip routing interface vlan 10  ip
address 192.168.10.1 255.255.255.0
ip helper-address 192.168.20.10  no
shutdown interface vlan 20  ip
address 192.168.20.1 255.255.255.0
```

```
no shutdown ! (repeat for VLANs 30,
40, 99)
```

```
! Routed uplink to R1 interface GigabitEthernet0/1
no switchport  ip address 10.0.0.2 255.255.255.252
no shutdown ! Trunk to SW-ACC-1 (repeat for Fa0/2,
Fa0/3, Fa0/4) interface FastEthernet0/1  switchport
trunk encapsulation dot1q  switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99  no
shutdown
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

5.2 Router R1

R1 is the edge gateway. It holds static routes for every VLAN subnet and sends them to SW-CORE, and it has a default route toward the external side.

```
hostname R1 enable secret <REDACTED>
service password-encryption interface
GigabitEthernet0/0  ip address 10.0.0.1
255.255.255.252  no shutdown ip route
192.168.10.0 255.255.255.0 10.0.0.2 ip
route 192.168.20.0 255.255.255.0 10.0.0.2
ip route 192.168.30.0 255.255.255.0
10.0.0.2 ip route 192.168.40.0
255.255.255.0 10.0.0.2 ip route
192.168.99.0 255.255.255.0 10.0.0.2 ip
route 0.0.0.0 0.0.0.0 GigabitEthernet0/1
```

5.3 Access switches

SW-ACC-1 to SW-ACC-4 use the same template: a trunk uplink (Fa0/1) to SW-CORE and an access port (Fa0/2) toward the department access point, assigned to the matching VLAN. Each switch also has a management address in VLAN 99. The example below shows SW-ACC-1, which serves VLAN 10.

```
hostname SW-ACC-1 enable secret <REDACTED>
service password-encryption vlan 10  name
ADMIN vlan 99  name MANAGEMENT interface
FastEthernet0/1  switchport mode trunk
switchport trunk allowed vlan 10,99  no
shutdown interface FastEthernet0/2
switchport mode access  switchport access
vlan 10  no shutdown interface vlan 99  ip
address 192.168.99.11 255.255.255.0  no
shutdown
ip default-gateway 192.168.99.1
```

5.4 Spanning Tree Protocol

Rapid PVST+ is used to prevent switching loops. SW-CORE is given priority 4096 so it is elected root bridge for every VLAN. Redundant paths stay blocked while remaining available as backup links, and in case of a link failure the protocol quickly activates an alternative path, which avoids broadcast storms and improves reliability.

```
spanning-tree mode rapid-pvst
```

```
spanning-tree vlan 1,10,20,30,40,99 priority 4096
```

5.5 DHCP service

A central DHCP server at 192.168.20.10 serves every VLAN. The **ip helper-address** command on each SVI forwards client broadcasts across VLAN boundaries to the server, which defines one pool per VLAN with the SVI as default gateway.

5.6 DNS service

The DNS service lets users reach resources by name instead of IP address. Three A records are defined.

Table 7. DNS records

Name	Type	Address
dhcp.corp	A	192.168.20.10
dns.corp	A	192.168.20.20
intranet.corp	A	192.168.20.30

5.7 Web service

Server-WEB (192.168.20.30) hosts the internal intranet portal at **http://intranet.corp**. The portal summarizes the features of the infrastructure and is reachable from all internal VLANs (Administration, IT and HR). Guest users are blocked by the BLOCK_GUEST ACL.

5.8 Wireless access

Each access point is wired to the access port of its department switch, so wireless clients inherit the VLAN of that port. The SSID therefore identifies the department, WPA2-PSK with AES protects the radio link, and the ACLs enforce the inter-department policy.

5.9 Security: access control lists

Two named extended ACLs implement the traffic policy. Both are applied **inbound** on the SVI of the source VLAN on SW-CORE.

BLOCK_GUEST prevents Guest users from reaching any internal network; they can only communicate with external resources.

```
ip access-list extended BLOCK_GUEST  deny ip
192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255  deny
ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.40.0 0.0.0.255 192.168.30.0
0.0.0.255  deny ip 192.168.40.0 0.0.0.255
192.168.99.0 0.0.0.255  permit ip any any interface
vlan 40
 ip access-group BLOCK_GUEST in
```

BLOCK_RH_ADMIN prevents HR staff from reaching the Administration network (192.168.10.0/24) while keeping access to other authorized resources such as the IT servers.

```
ip access-list extended BLOCK_RH_ADMIN  deny ip
192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
permit ip any any interface vlan 30  ip access-group
BLOCK_RH_ADMIN in
```

6. Testing and Validation

The validation follows three ordered phases: baseline connectivity first, service availability second and security enforcement last. Checking each phase before the next one makes it possible to attribute any failure to its cause: for example, a blocked ping is only meaningful if the same ping succeeded before the ACL was applied.

6.1 Baseline connectivity (before ACLs)

Before the ACLs were applied, communications between all VLANs were verified. The ping tests confirmed that routing was operational and that DHCP distributed addresses correctly in each VLAN. Round-trip times were consistently below 65 ms, which confirms normal operation.

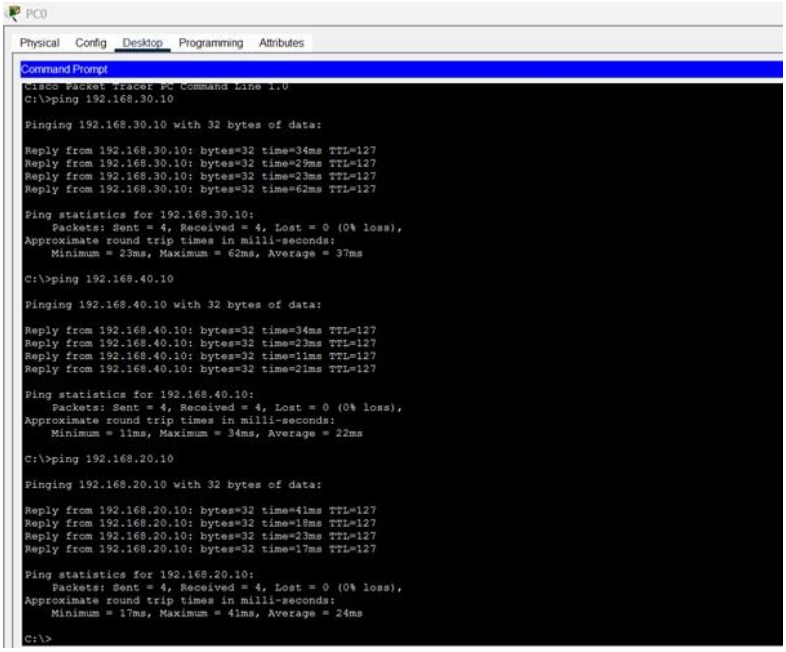


Figure 2. Ping tests from an Administration PC before the ACLs: 0% loss toward the HR, Guest and IT hosts.

Table 8. Baseline results (from the Administration PC)

Destination	Packets sent / received	Loss	Round-trip time (ms)
192.168.30.10 (HR host)	4 / 4	0%	min 23, max 62, avg 37
192.168.40.10 (Guest host)	4 / 4	0%	min 11, max 34, avg 22
192.168.20.10 (IT server)	4 / 4	0%	min 17, max 41, avg 24

6.2 Service availability

Users in all internal VLANs (Administration, IT and HR) successfully reached the intranet portal at http://intranet.corp with the Packet Tracer web browser. The page loaded correctly and displayed the summary of the enterprise network infrastructure, which also confirms that DNS resolution works.

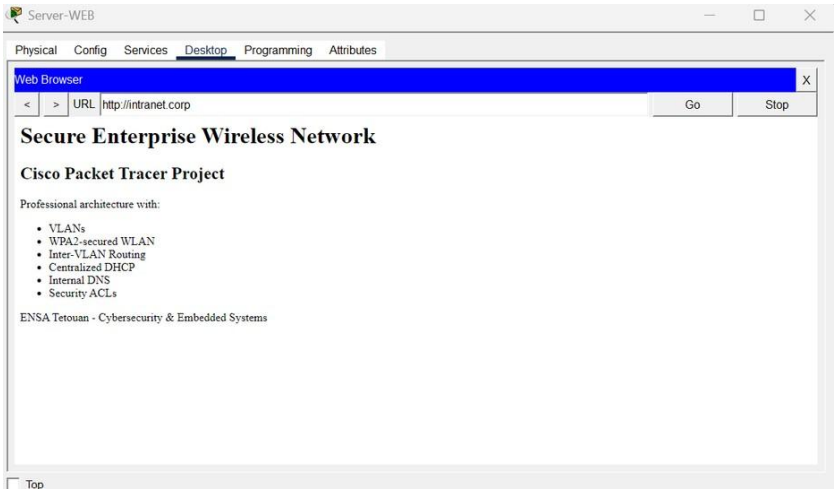


Figure 3. Intranet portal at http://intranet.corp, loaded from an internal VLAN.

6.3 Security enforcement (after ACLs)

After both ACLs were applied, the following restrictions were confirmed:

- Guest users (VLAN 40) pinging internal servers all returned 100% packet loss (request timed out), which confirms BLOCK_GUEST.
- HR users (VLAN 30) trying to reach the Administration network (192.168.10.x) were all blocked with 100% packet loss, which confirms BLOCK_RH_ADMIN.
- HR users could still reach the IT servers (192.168.20.x) with 0% packet loss, which confirms that the ACL only removes the targeted traffic.
- Guest users could not load the intranet portal: the HTTP connection from VLAN 40 failed.

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (1), with S
show access-lists
Extended IP access list BLOCK_GUEST
    10 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
    20 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
    30 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
    40 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
    50 permit ip any any
Extended IP access list BLOCK_RH_ADMIN
    10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
    20 permit ip any any

SW-CORE#
```

Figure 4. Output of show access-lists on SW-CORE, listing both ACLs and their sequence numbers.

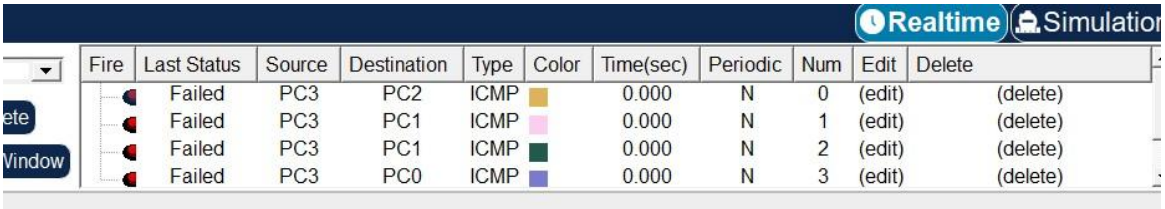


Figure 5. Simulation mode: ICMP packets from the Guest PC (PC3) toward PC0, PC1 and PC2 report Failed.

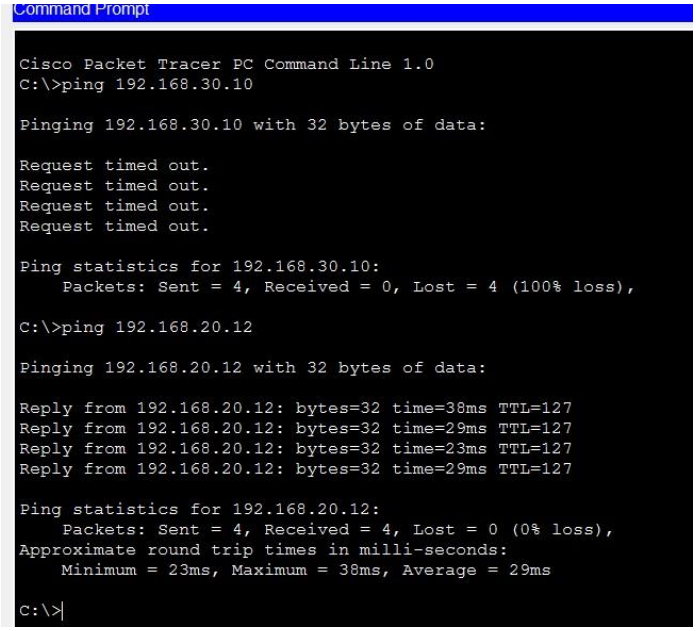


Figure 6. Console test from the Administration PC: the restricted path times out (100% loss) while the path to the IT network succeeds (0% loss).

Stateless behaviour. The console test above pings an HR host from an Administration PC. The request reaches HR, but BLOCK_RH_ADMIN denies the HR reply on its way back to the Administration network, so the ping times out. This illustrates that ACLs do not track connections. The ping toward the IT host is unaffected, which confirms the policy is targeted.

6.4 Summary of results

Table 9. Test campaign summary

Phase	Scope	Result
1. Baseline connectivity	DHCP leases and inter-VLAN pings, RTT below 65 ms	Passed
2. Service availability	DNS resolution and intranet portal from internal VLANs	Passed
3. Security enforcement	Guest isolation, HR to Administration block, HR to IT allowed, Guest web access blocked	Passed

7. Security Analysis

The design combines several layers of defence: VLAN segmentation at Layer 2, ACLs at Layer 3, WPA2 encryption on the radio link, protected privileged access on the devices (enable secret and password encryption) and a dedicated management VLAN.

Table 10. Resulting policy by source VLAN

From	Administration	IT / servers	HR	Guest	External
Administration	-	Allowed	Reply blocked	Reply blocked	Allowed
IT	Allowed	-	Allowed	Reply blocked	Allowed
HR	Denied	Allowed	-	Reply blocked	Allowed
Guest	Denied	Denied	Denied	-	Allowed

"Reply blocked" means the request reaches the destination but the destination's answer is dropped by an ACL, so the session cannot be established. The matrix is derived from the ACL logic; the cases covered by tests are listed in Section 6.

Observations that a production design should address:

- **Management VLAN:** only Guest is denied access to VLAN 99. Administration, IT and HR are not restricted by an ACL.
- **Native VLAN mismatch:** the switches log CDP native VLAN mismatch warnings on some links. The native VLAN should be aligned on both ends of every trunk, and the physical cabling verified.
- **Wireless authentication:** WPA2-PSK relies on a shared key per SSID; 802.1X with RADIUS would provide per-user authentication.
- **Layer 2 attacks:** DHCP Snooping and Dynamic ARP Inspection are not enabled on the access switches.
- **Guest name resolution:** the guest policy denies VLAN 20, which hosts the internal DNS server, so guests need a separate resolver for Internet names.
- **Credentials:** lab credentials are simple; production devices need strong secrets and SSH-only remote access.

8. Limitations and Future Work

Limitations. All results come from a simulator: latency values are simulated and the tests use ICMP and HTTP only. Wireless propagation, roaming and radio interference are not modelled realistically, and the design has no redundancy at the core or edge.

Future work to move toward a production-grade design, in suggested order:

- Resolve the native VLAN mismatch warnings on the trunk links.
- Enable DHCP Snooping and Dynamic ARP Inspection on the access switches.
- Deploy a wireless LAN controller (WLC) with a RADIUS server to migrate to WPA2-Enterprise and obtain seamless roaming between access points.
- Restrict access to the management VLAN and enable SSH-only administration.
- Add port security and BPDU protection on access ports.
- Introduce a stateful firewall at the edge, and redundancy at the core and uplink.

9. Conclusion

This project demonstrates the design, implementation and validation of a secure enterprise wireless network with Cisco Packet Tracer. The main components of an enterprise network were implemented: VLAN segmentation, inter-VLAN routing through Layer 3 switching, centralized DHCP, DNS and HTTP services, WPA2-secured wireless access, and layered security through ACLs.

The architecture reflects good industry practice: departmental traffic is separated with VLANs, shared services are centralized in the IT VLAN, the principle of least privilege is applied through ACLs, and dedicated access points are used instead of consumer Wi-Fi routers. The test approach, which validates baseline connectivity, service availability and security enforcement separately, mirrors the methodology used in real enterprise deployments.

The improvements identified for a production environment (native VLAN alignment, DHCP Snooping and Dynamic ARP Inspection, and a wireless controller with RADIUS for WPA2-Enterprise and seamless roaming) provide a clear path from this validated simulation toward a deployable design.

Appendix A. Technical Skills Demonstrated

Table 11. Skills and technologies applied

Skill area	Technologies and concepts
Enterprise network design	Three-tier hierarchical topology, Cisco 2911, 3560 and 2960
Cisco switching	802.1Q trunks, access ports, SVI configuration
VLAN segmentation	VLAN creation and naming, inter-VLAN isolation
Wireless LAN	WPA2-PSK, AES encryption, per-department SSIDs, access points
Inter-VLAN routing	Layer 3 switching with ip routing, SVI gateways
DHCP management	Centralized DHCP server, ip helper-address relay, DHCP pools
DNS administration	A records, access by host name
Web services	HTTP intranet portal deployment and access control
Network security (ACL)	Named extended ACLs, inbound filtering, traffic policy
Spanning Tree	Rapid PVST+, root bridge election, loop prevention
Simulation and testing	Cisco Packet Tracer, systematic validation methodology

Appendix B. Project Resources

The Packet Tracer simulation file, the sanitized CLI configurations and the detailed documentation (architecture, addressing, security implementation and test plan) are published in the project repository. The configurations contain no passwords or pre-shared keys.